

Association between adherence to the Mediterranean diet and metabolic syndrome in children and adolescents: a systematic review and meta-analysis

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ABSTRACT

Background The Mediterranean diet has been associated with improved metabolic health among adults. This meta-analysis aimed to evaluate the association between adherence to a Mediterranean diet and metabolic syndrome among children and adolescents.

Methods A systematic review of the literature was conducted by searching PubMed, the Excerpta Medica Database (Embase, via OvidSP), Scopus and Web of Science, covering publications from inception to April 2024. Studies were included if they evaluated the association between adherence to a Mediterranean diet and metabolic syndrome among children and adolescents. A random-effects model was used to estimate the summary effect size.

Results A total of eight studies involving children and adolescents aged 6 to 19 years were included in the meta-analysis (n=6562, 50% girls). Low adherence to the Mediterranean diet was associated with greater odds of metabolic syndrome than was medium/high adherence (odds ratio [OR] = 2.18, 95% confidence interval [CI] 1.45 to 3.28; $p < 0.001$; inconsistency index [I^2] = 64.15%).

Conclusion This meta-analysis suggested that increased adherence to a Mediterranean diet is associated with lower probabilities of having cardiometabolic risk among children and adolescents. Thus, our findings suggest that public health strategies are needed to implement and promote effective actions toward healthy eating habits in children and adolescents.

INTRODUCTION

Adolescence and childhood are critical periods for the development of metabolic health,¹ as there are times when changes in diet and lifestyle can have long-term effects on health outcomes.² Poor metabolic health in adolescence can lead to the development of chronic diseases such as type 2 diabetes, cardiovascular disease and obesity in adulthood.^{3–6} The prevalence of metabolic disorders is increasing globally, especially among adolescents, and this has become a public health concern.^{2,7}

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ The Mediterranean diet (MedDiet) has been associated with improved metabolic health outcomes, particularly in adult populations.
- ⇒ Metabolic syndrome (MetS) in childhood and adolescence is an emerging global public health issue, with increasing prevalence and long-term health implications.
- ⇒ Previous studies in youth populations have shown inconsistent associations between MedDiet adherence and MetS risk.

WHAT THIS STUDY ADDS

- ⇒ This is the first systematic review and meta-analysis exclusively focused on the association between adherence to the MedDiet and MetS in children and adolescents.
- ⇒ Low adherence to the MedDiet is significantly associated with increased odds of having MetS among youth (odds ratio [OR] = 2.18; 95% confidence interval [CI]: 1.45 to 3.28).
- ⇒ The findings were consistent across diverse countries and dietary assessment tools, despite moderate heterogeneity.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ These results support the implementation of public health initiatives aimed at promoting Mediterranean-like dietary patterns in early life stages.
- ⇒ The findings highlight the importance of incorporating MedDiet education and interventions into school- and community-based health promotion programs.
- ⇒ Future research should prioritize longitudinal studies to assess causality and inform evidence-based dietary guidelines for paediatric populations.

The main features of metabolic health include ideal levels of blood sugar, triglycerides, high-density lipoprotein (HDL) cholesterol, blood pressure and waist circumference.^{8–11} Abnormalities in these metabolic health levels can lead to obesity, hypertension,

hyperlipidaemia and insulin resistance.^{4 12 13} The key contributing factors include diet and behaviour, such as an increased intake of foods containing high amounts of saturated fat and carbohydrates, particularly sweetened beverages; a lack of physical activity; and smoking.^{1 7 9 14} The World Health Organization (WHO) recognises the significant impact of diet on disease risk and has developed a Global Action Plan to address unhealthy dietary patterns.¹⁵ The plan aims to reduce behavioural risk factors such as physical inactivity, tobacco use, harmful alcohol consumption and unhealthy eating habits. The WHO recommends various dietary changes to promote better health, including maintaining balanced energy intake, limiting saturated and trans fats, transitioning to unsaturated fats, increasing fruit and vegetable intake, and reducing sugar and salt consumption. These recommendations are often found in regional diets, such as the Mediterranean diet (MedDiet), which is characterised by high consumption of fruits, vegetables, whole grains, legumes, nuts, fish, and olive oil and low consumption of red and processed meats, sweets and refined grains.¹⁶

A MedDiet has been associated with improved metabolic syndrome (MetS) outcomes, including a decreased risk of type 2 diabetes, cardiovascular disease and obesity, among adults.¹⁷ Several studies have associated the MedDiet with individual MetS-related factors (blood sugar, triglycerides, HDL cholesterol, blood pressure and waist circumference).^{14 17–19} A recent meta-analysis highlighted that interventions based on a MedDiet are highly effective in lowering childhood and adolescent obesity and body mass index.²⁰ However, the extent to which different levels of adherence to the MedDiet are associated with MetS outcomes in youth remains poorly understood. While several studies have reported associations, the consistency and strength of these findings across varying adherence categories warrant further synthesis and evaluation.

Several studies have investigated the associations between adherence to a MedDiet and MetS outcomes among adolescents,^{21–26} but the results have been inconsistent. Some studies have shown a positive association between adherence to the MedDiet and improved MetS risk,^{21–26} whereas others have reported no significant association.²⁷ To our knowledge, no previous systematic review or meta-analysis has synthesised the available evidence examining the association between adherence to the MedDiet and MetS among children and adolescents. Therefore, the aim of this systematic review and meta-analysis was to investigate whether adherence to a MedDiet is associated with MetS in this population.

METHODS

We followed the methods suggested in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.²⁸ In accordance with the guidelines, our systematic review/meta-analysis protocol was registered with the International Prospective

Register of Systematic Reviews on 11 July 2022 and was last updated on 19 January 2023 (registration number CRD42022343383).

Eligibility criteria

To be eligible for inclusion in the present review, studies needed to meet the following criteria: (i) were children or adolescents aged 5–19 years, all countries and ethnic backgrounds; (ii) had a MetS outcome; (iii) had an adherence to the MedDiet assessed by a validated questionnaire; and (iv) had a longitudinal and cross-sectional study design. However, to narrow the scope of the meta-analysis, only studies that associated the MedDiet with the outcome of MetS were included. In contrast, the exclusion criteria included adults and adolescents/children with evidence of metabolic or autoimmune liver disease, diabetes or the presence of known systemic disease. Studies were also excluded if they presented duplicate data from the same source and year. When two studies used duplicate data from the same source and year, the study with the larger sample size was selected.

Information sources

Two authors independently conducted systematic searches in the following databases: Excerpta Medica Database (Embase, via OvidSP), Scopus, Web of Science and PubMed. The searches included studies published from database inception to 22 April 2024. The search was limited to peer-reviewed studies published in English. A professional librarian was consulted to ensure the quality of the search strategy.

Search strategy

A search strategy was used to identify all eligible studies published before 22 April 2024. The search strategy included controlled vocabulary terms specific to each database along with keywords derived from scoping index articles relevant to this meta-analysis. The following terms related to the MedDiet were used: MEDdiet, Mediterranean Diet Quality Index for Children and Adolescents (KIDMED) score, MedDiet score, MedDiet and hypertension, MedDiet and diabetes, KIDMED and MedDiet score. The following outcome terms related to MetS were included: MetS or components (hyperlipidaemia, hypertriglyceridaemia, insulin resistance, hyperglycaemia, hypertension) combined with health. The reference lists of eligible studies were manually examined to find further relevant papers and included if suitable. Any disagreements were settled through consensus with a third author. The complete search strategies used for all the databases are presented in online supplemental file 2.

Selection process

Two authors (VPD and AG-H) thoroughly examined the entire texts of identified publications for eligibility after removing duplicates and systematically reviewed the titles and abstracts of eligible manuscripts. In case of discrepancies or uncertainty regarding eligibility, a third reviewer (CA-S) was consulted to reach consensus.

Table 1 Results of the critical appraisal of the included studies. +, self-reported outcomes; ++, outcome with independent blinded observers

Author (year)	Study design	Selection			Comparability		Outcome	
		Representativeness of the sample	Sample size	Non-respondents	Ascertainment of exposure	Based on design and analysis	Assessment of outcome	Statistical test
Arenaza <i>et al</i> (2019) ²¹	Cross-sectional	+	+			+	+	+
Bibiloni <i>et al</i> (2011) ²²	Cross-sectional	+	++				+	+
George <i>et al</i> (2021) ²³	Cross-sectional	+				+	+	+
Martino <i>et al</i> (2016) ²⁴	Cross-sectional	+					+	+
Mirzaei <i>et al</i> (2022) ²⁵	Cross-sectional	+				+	+	+
Mohammadi <i>et al</i> (2022) ²⁶	Cross-sectional	+	+			+	+	+
Kim <i>et al</i> (2021) ²⁷	Cross-sectional	+	+	+			+	+
Yilmaz <i>et al</i> (2021) ³¹	Cross-sectional	+	+			+	+	+

Data collection process and data items

The following information was extracted from the included studies: (a) reference (year); (b) country; (c) characteristics of the subjects, including total sample size, sex ratio, age range and mean age; (d) type of questionnaire used; and (e) outcome information, including information on MetS and adherence to the MedDiet. All results from each study that were compatible with each outcome domain were included. In cases where missing information was identified in the published paper, the relevant authors of those studies were contacted directly to request the missing information.

Risk of bias assessment

The extracted data were systematised via a specially constructed standardised data extraction form (see study protocol), and the extracted data were then reviewed for consistency by the reviewers. Each included study was independently reviewed for critical appraisal and bias reduction, and a modified version of the Newcastle Ottawa cohort scale was used for cross-sectional research (online supplemental file 1).²⁹ Consensus was used to settle disagreements; the results are shown in table 1. General study information, participant and intervention characteristics, withdrawal information and outcome information were extracted.

Effect measures

The effect size of each study was evaluated using the odds ratio (OR) and its 95% confidence interval (CI). All studies were included in the meta-analysis to assess the association between adherence to the MedDiet and MetS among children and adolescents. All analyses were conducted using Stata version 17.0 (StataCorp, College Station, TX, USA).

Strategy for data synthesis

A meta-analysis was also conducted to investigate the associations between MetS and MedDiet adherence. To determine the overall estimated OR and its related 95% CIs, the random-effects inverse variance model with the traditional DerSimonian and Laird method was used. To measure the percentage of disparities across studies, we used the inconsistency index (I^2), which was calculated with the Cochran Q statistic. I^2 values of 0%–40%, 30%–60%, 50%–90% and 75%–100% were considered cut-off values for non-important, moderate, substantial and considerable degrees of inconsistency, respectively.³⁰

Assessment of publication bias

Publication bias was assessed via a funnel plot, and Egger's test was used for the assessment of bias.

RESULTS

Study selection

A total of eight cross-sectional studies were included in the meta-analysis. We identified 3775 articles by searching the databases MEDLINE (via PubMed), Embase (Ovid) and Web of Science. A total of 1456 studies were excluded because of duplication, while 1565 were classified as ineligible by automation tools. Another 546 studies were removed for additional reasons. We examined the titles and abstracts of 208 studies and excluded 157 on the basis of our findings. We screened 51 articles for eligibility. Eight studies were included in the meta-analysis. The PRISMA flow chart is shown in figure 1.

Study characteristics

The characteristics of the eight studies included in this meta-analysis are presented in table 2. All studies were

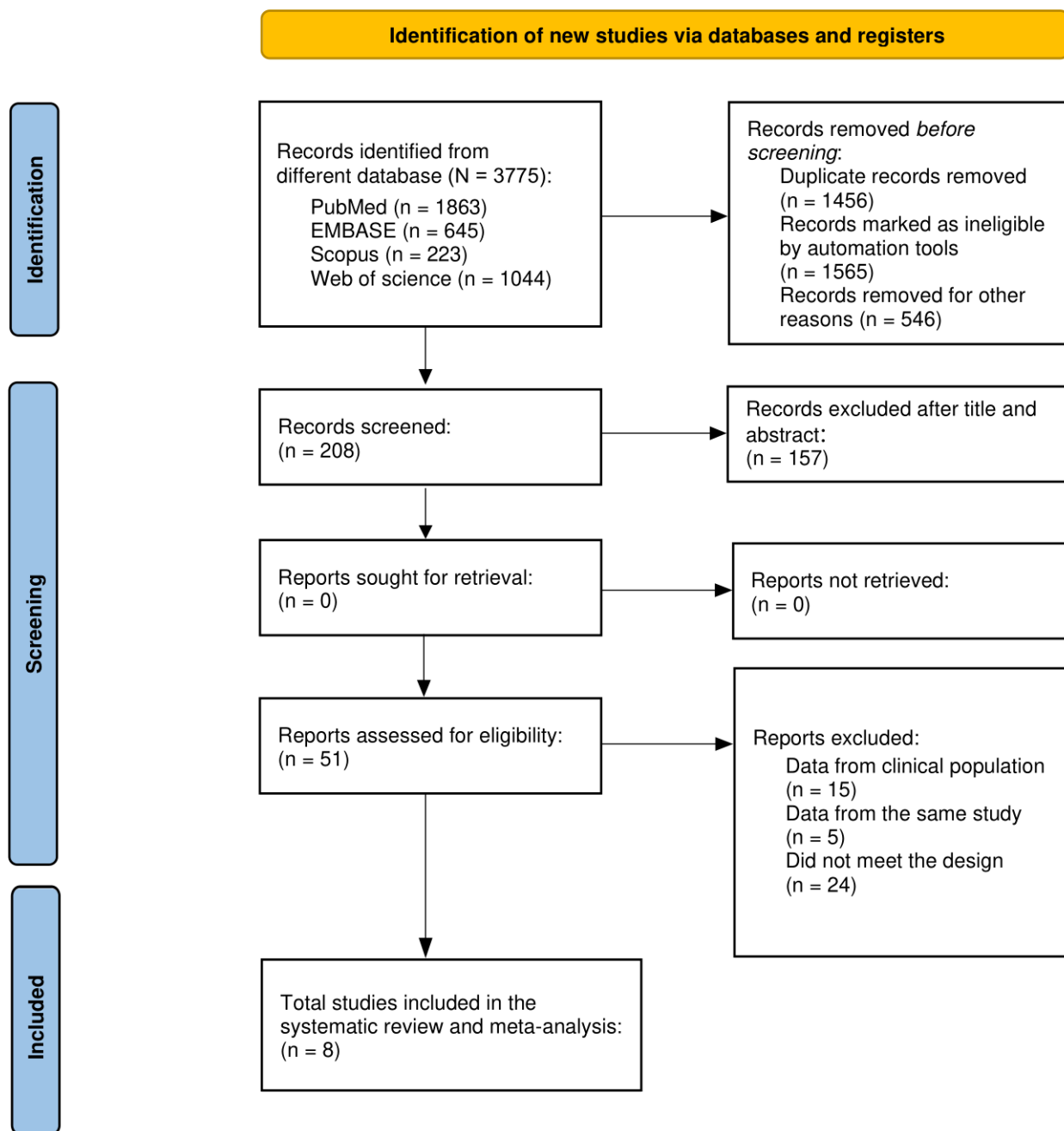


Figure 1 Preferred Reporting Items for Systematic Reviews and Meta-Analyses flowchart.

cross-sectional in design and published between 2011 and 2022. The studies were conducted in diverse geographic settings, including Europe (Spain, Greece, Italy), the Middle East (Iran), the USA and Turkey. The sample sizes ranged from 137 to 1972 participants, with the proportion of female participants ranging from 47.5% to 60.5%.

There was considerable heterogeneity in the age of participants across studies, ranging from 6 to 19 years, covering both early and late adolescence. Dietary adherence to the MedDiet was assessed via different tools: five studies employed the KIDMED index, whereas three used FFQs to derive MedDiet scores or patterns. The KIDMED is a validated, structured index specifically designed to measure

adherence to the MedDiet in young populations and is based on a 16-item scoring system. In contrast, FFQs are broader tools used to assess usual dietary intake, from which MedDiet adherence scores are derived using predefined criteria. Similarly, outcome measures varied, with most studies using International Diabetes Federation (IDF) criteria for defining MetS, while some also included definitions based on homeostatic model assessment for insulin resistance (HOMA-IR) or other phenotype-based approaches (eg, metabolically unhealthy obesity [MUO]). The main results consistently indicated an inverse association between adherence to the MedDiet and the presence of MetS or its components, although

Table 2 Characteristics of the studies included in the meta-analysis

Reference (year)	Country	Sample size (% girls)	Age range	Tool	Outcomes
Arenaza <i>et al</i> (2019) ²¹	Europe (multi centre)	137 (48.9%)	12.5–17.5	MDS	Lower number of participants with high adherence to the MedDiet is related to higher OR of MetS (OR=2.13; 95% CI 1.03 to 4.40).
Biblioni <i>et al</i> (2011) ²²	Spain	362 (60.5%)	12–17	FFQ	Higher adherence to MedDiet was associated with significantly lower OR of having MetS, but half of the adolescents showed high adherence to MedDiet.
George <i>et al</i> (2021) ²³	Greece	1972 (49.8%)	9–13	KIDMED	Poor adherence to the MedDiet was associated with an increased likelihood for central obesity (OR=1.31; 95% CI 1.01 to 1.73), hypertriglyceridemia (OR=2.80; 95% CI 1.05, 7.46), and insulin resistance (OR=1.31; 95% CI 1.05 to 1.64).
Martino <i>et al</i> (2016) ²⁴	Italy	1643 (47.5%)	6–14	KIDMED	Poor or medium adherence to the MedDiet had higher prevalence of MetS compared to high adherence (OR=1.9).
Mirzaei <i>et al</i> (2022) ²⁵	Iran	203 (50.2)	12–18	FFQ	High adherence to Mediterranean-like dietary pattern was related to 72% (OR=0.28; 95% CI 0.10 to 0.76) and 67% (OR=0.33; 95% CI 0.11 to 0.98) lower odds for having MUO based on IDF and HOMA-IR definitions, respectively.
Mohammadi <i>et al</i> (2022) ²⁶	Iran	203 (50.2)	12–18	FFQ	MUO phenotype in the third tertile of MedDiet score was lower than in the first tertile (60.9% vs 21.0%). A similar result was seen when the IDF/HOMA-IR definition was considered (51.6% vs 19.4%).
Kim <i>et al</i> (2021) ²⁷	USA	1675 (50%)	12–19	KIDMED	Using the KIDMED score as a continuous variable also showed no observed association with MetS (OR=1.01, 95% CI 0.90 to 1.12). Furthermore, comparing individuals with average adherence to the MedDiet to those with poor adherence yielded an OR of 0.86 (95% CI 0.48 to 1.54), indicating no significant difference ($p=0.61$).
Yilmaz <i>et al</i> (2021) ³¹	Turkey	367 (52.3%)	10–18	KIDMED	A significant proportion of subjects with metabolic syndrome (51.2%) demonstrated poor adherence to the MedDiet.

FFQ, food frequency questionnaire; HOMA-IR, Homeostatic Model Assessment of Insulin Resistance; IDF, International Diabetes Federation; KIDMED, Mediterranean Diet Quality Index in children and adolescents; MDS, Mediterranean diet score; MedDiet, Mediterranean diet; MetS, metabolic syndrome; MUO, metabolically unhealthy obesity; p , statistical significance.

one study did not find a statistically significant association.

Results of individual studies

The results of the individual studies are presented in figure 2. Children and adolescents with low adherence to the MedDiet had greater odds of developing MetS compared to those with medium to high adherence. The OR for this association was 2.18, with a 95% CI ranging from 1.45 to 3.28 ($p < 0.001$). The inconsistency among the studies, as assessed by the I^2 statistic, was 64.15%.

Results of the publication bias analysis

We could not observe any publication bias in these studies. The funnel plot was not asymmetrical, and Egger's test yielded a p value of 0.162 (figure 3).

Results of the sensitivity analysis

A sensitivity analysis was conducted by sequentially omitting one study at a time to assess the robustness of the overall effect estimate. The pooled ORs remained stable across all iterations, ranging from 1.91 to 2.43, and all remained statistically significant ($p \leq 0.001$). These findings indicate that no single study unduly

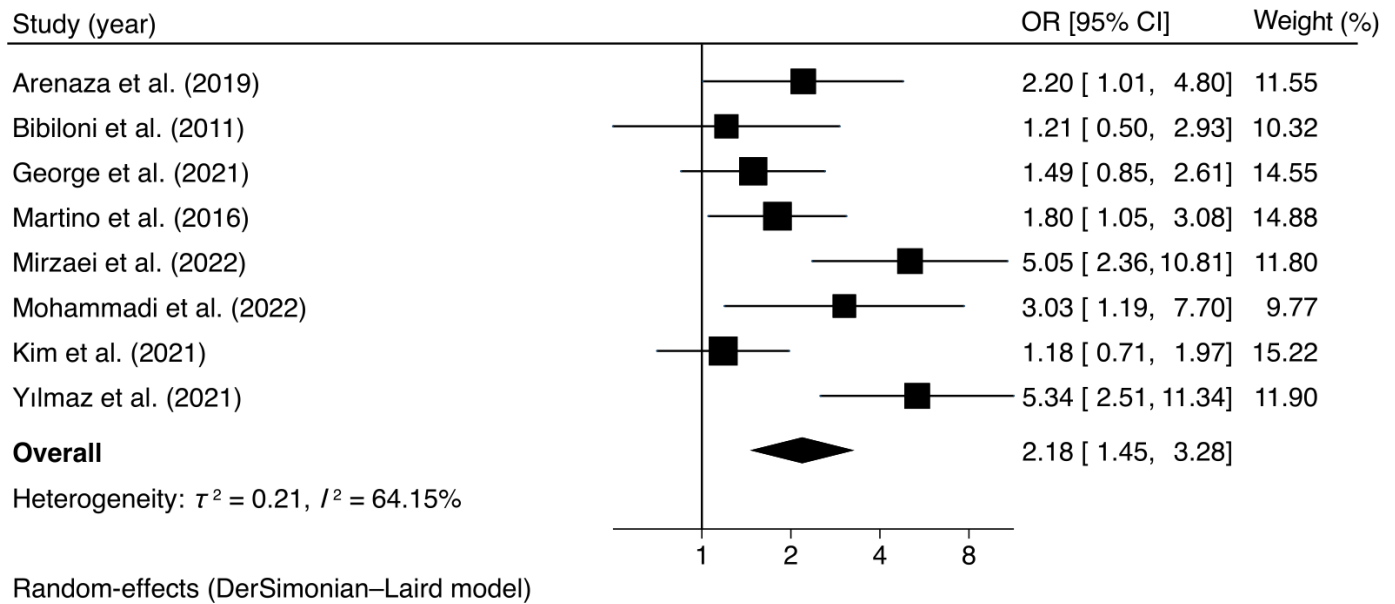


Figure 2 Results of the individual studies. τ^2 , heterogeneity; I^2 , inconsistency index.

influenced the overall results, supporting the robustness of the association between low adherence to the MedDiet and higher odds of MetS in children and adolescents (figure 4).

DISCUSSION

The association between adherence to the MedDiet and MetS among children and adolescents has been a topic of increasing interest in recent years. Our study revealed

that children and adolescents with lower MedDiet adherence had a greater risk of MetS than did those with higher adherence, providing important insights into the potential benefits of the MedDiet for MetS in children and adolescents. Specifically, higher adherence to the MedDiet was associated with lower odds of MetS, which includes a clustering of metabolic abnormalities such as abdominal obesity, high blood pressure, dyslipidaemia and impaired glucose metabolism. Overall, our

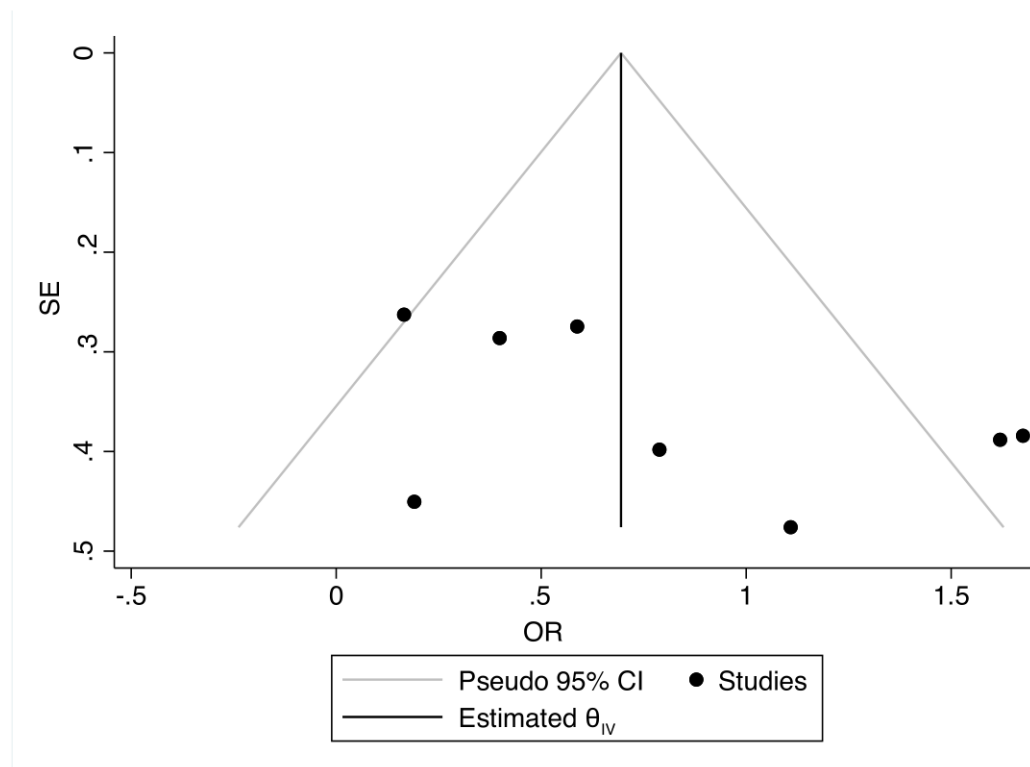


Figure 3 Funnel plot. CI, confidence interval; OR, odds ratio; SE, standard error.

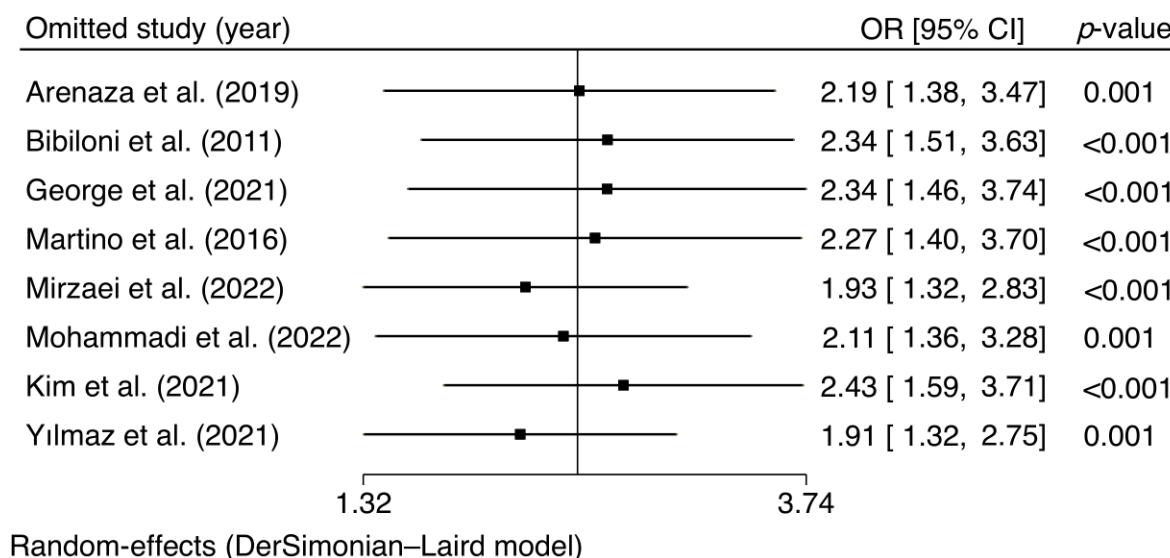


Figure 4 Sensitivity analysis. CI, confidence interval; OR, odds ratio.

meta-analysis included a total of eight studies with a combined sample size of 6562 children and adolescents. The studies were conducted in various countries, such as Spain,²² Greece,²³ Italy,²⁴ Iran,^{25 26} the USA²⁷ and Turkey³¹ and included cross-sectional studies. However, it should be noted that there was considerable inconsistency among the studies included in our meta-analysis. This variability encompassed differences in study design (cross-sectional vs longitudinal), participant characteristics (eg, age range from early childhood to late adolescence, sample size and geographic region) and tools used to assess adherence to the MedDiet. Some studies have used the KIDMED, a validated instrument specifically developed for children and adolescents, whereas others have employed FFQs and derived MedDiet adherence scores on the basis of predefined criteria. Additionally, definitions of MetS vary across studies, with some applying IDF criteria and others using alternative classifications. These methodological and population differences likely contributed to the observed inconsistency and should be considered when the pooled effect estimates are interpreted.

These findings are consistent with previous research in adults, suggesting that adherence to the MedDiet may have favourable effects on MetS outcomes.¹⁷ However, unlike the study performed by Bakaloudi *et al*,¹⁷ our study focused only on children and adolescents, presenting a well-designed systematic review and meta-analysis of paediatric populations. The results of this systematic review and meta-analysis are important from the public health perspective for this specific population since, during the process of growth, various nutritional needs must be met to sustain constant body changes.³² In addition, healthy behaviours play a key role at early ages and must be promoted, as healthy eating behaviours can extend from childhood into adulthood,³² helping individuals avoid and prevent obesity and health-related problems.

Several potential mechanisms may explain the observed association between adherence to the MedDiet and

improved MetS.^{11 25 33} The MedDiet is characterised by a high intake of fruits, vegetables, whole grains, legumes, nuts, seeds and healthy fats (such as olive oil and fish) and a moderate intake of dairy products, poultry and eggs, with limited intake of red meat and processed foods.^{34 35} The high intake of fruits and vegetables, whole grains and legumes provides important sources of dietary fibre, vitamins, minerals and antioxidants, which may help reduce inflammation, oxidative stress, and insulin resistance and improve insulin sensitivity.^{18 36} Healthy fats in the MedDiet, such as monounsaturated fats from olive oil and omega-3 fatty acids from fish, may also have beneficial effects on lipid metabolism, blood pressure regulation and glucose homeostasis.³⁶ In addition, the MedDiet is typically low in added sugars and highly processed foods, which may contribute to improved MetS outcomes.

Limitations of the present systematic review include the fact that the degree of adherence to the MedDiet could be subject to regional variations, potentially introducing further biases. For example, individuals with high adherence to the MedDiet living in Mediterranean areas may consume certain foods more frequently than may those residing in non-Mediterranean regions. Furthermore, the need to convert data for the purpose of quantitative analysis presents an additional limitation to our study; additionally, we restricted our inclusion criteria to studies published in English. In addition, most of the included studies were cross-sectional, which limits the ability to establish causality or determine the temporal relationship between adherence to the MedDiet and MetS outcomes. However, further well-designed prospective cohort studies and randomised controlled trials are needed to confirm the findings of our meta-analysis and elucidate the underlying mechanisms involved. All the studies used a cross-sectional design, so the observed results should be interpreted with caution. Another significant limitation is that studies use different questionnaires to determine adherence to the MedDiet, as well as different points for

comparisons between categories. Additionally, there is no single criterion for defining MetS in the paediatric population, so the observed inconsistency could be due to these limitations. Finally, a formal assessment of the certainty of evidence (eg, using the Grading of Recommendations Assessment, Development and Evaluation approach) was not performed, as this was beyond the scope of the present review. Conversely, this study has certain strengths. For example, to our knowledge, this is the first systematic review and meta-analysis that examined the association between adherence to a MedDiet and MetS among children and adolescents. In addition, our study included studies that employed validated MedDiet adherence scores to assess adherence to the MedDiet, thereby enhancing the accuracy of our findings. Moreover, our study did not reveal any publication bias.

In conclusion, the MedDiet could be a protective factor against MetS among children and adolescents. These findings highlight the potential benefits of the MedDiet for promoting MetS early in life and may have important implications for public health policies and interventions aimed at improving the dietary patterns of children and adolescents. Further research is warranted to better understand the underlying mechanisms and long-term effects of adherence to the MedDiet on MetS in this population.

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Contributors VPD and AGH were responsible for the literature search; JFLG and CV participated in the data extraction; VPD, AGH and CAS were responsible for the study quality assessment; VPD, JFLG, AR and CV were involved in the study design and data analysis; VPD, AGH, and JFLG drafted the manuscript; JFLG and AR participated in the critical review of the manuscript; CV, JFLG, FQC and CAS were responsible for the critical revision of the manuscript; CV and CAS were responsible for the study conception and study supervision; all the authors were involved in the data contribution, data interpretation and approval of the manuscript; and all the authors approved the final draft of the manuscript. VPD is the guarantor of the article.

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